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Experimental Studies on the Pathogenesis and Histogenesis of Ovarian Tumors in Mice*

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The role of the anterior pituitary in the regulation of the gonadal function, and the reciprocal influence of the gonads on the pituitary gland have been established (81, 93). Gonadotrophic and ovarian hormones, under normal circumstances, maintain a state of antagonistic interaction (60). It is not difficult to understand, however, that certain conditions may arise that disturb the usual secretory activity of one or more of these endocrine glands thus resulting in hormonal imbalances. The prolonged exposure of animals of some species to estrogens results in the appearance of atypical growths or cancers in some genital tissues and in other organs (31). The nature of the neoplastic action of estrogens, whether estrogens act directly on the body tissues or indirectly through the disturbance of the production of pituitary hormones or otherwise, still awaits the results of further study.

The influence of pituitary hormones on the development of tumors has been studied by the use of animals given injections of different pituitary preparations, and by observing the effect of hypophysectomy on tumor-bearing animals or animals given carcinogenic substances. These investigations so far give little conclusive information (79). Recently, Pfeiffer and Hooker (69) obtained small local overgrowths of testicular interstitial cells in mice of the A strain that were given daily injections of pregnant mare's serum for several months. These growths resembled early stages in the development of the interstitial cell tumors that have been induced in estrogen-treated mice of this strain. The difficulty of obtaining purified and

standardized gonadotrophic hormones, and the formation of antihormones have impeded adequate study of the long-term effects of gonadotrophins.

The present experiments are based on two principles: (a) the capability of the liver to inactivate ovarian hormones, *i.e.* estrogen and progesterone, when the hormones circulate through the hepatic portal system (6, 17, 35, 45, 46, 55, 77, 78, 91); and (b) the increase of intrinsic gonadotrophic hormones subsequent to castration as determined by bioassay of urinary, blood and hypophyseal gonadotrophins, and in experimental parabiosis of an intact with a castrated or roentgen-rayed animal (20, 21, 23, 28, 54). A condition of endocrine imbalance might be produced by the transplantation of ovaries into spleens of castrated mice. Such conditions should permit the study of continuous actions of endogenous pituitary gonadotrophins in ovarian tumorigenesis. Biskind and Biskind (4) reported that tumors developed in ovaries transplanted into spleens of 3 castrated female rats. Lipschütz and associates (52) observed only growth of lutein cells in some female guinea pigs with intrahepatic and intrasplenic grafts of ovaries.

Following the exposure of immature rats to roentgen irradiation (459 r or 574 r) directed to abdominal fields overlying the ovaries, Drips and Ford (18) observed hyperplasia and ingrowth of the germinal epithelium as well as the development of atypical, hyperplastic structures of epithelial cells that assumed an acinar arrangement and resembled "carcinomatous ovarian cystadenomata of the ovary of human beings." They found that such hyperplastic follicular structures occurred 4 months after irradiation, but more extensive growth had taken place in the rats that were killed 8 months later. The formation of granulosa-cell tumors, luteomas, and "tubular

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adenomas" in mice subsequent to similar applications of roentgen rays has been reported by Furth (29) and others (34, 85), and the histogenesis of the ovarian tumors has also been studied by several workers.

The present investigation is a study of the pathogenesis and the possible histogenesis of the ovarian tumors appearing in castrated mice bearing the intrasplenic ovarian transplants. A preliminary note on these tumors has been reported elsewhere (50).

MATERIALS AND METHODS

Inbred mice of the Strong A and C3H strains,¹ and hybrid mice (AC₃, AC₅)² were used. They were kept in an air-conditioned room and fed a commercial diet (Nurishmix) and water with occasional supplements of grain (a mixture of wheat, oats, and sunflower seeds) and calf-meal pellets. Six groups of experiments were set up (Table I): (a) 21 castrated male mice with homotransplants of an ovary into the spleen; (b) 52 castrated female mice with autotransplants of an ovary into the spleen; (c) 25 castrated female mice with subcutaneous autotransplants of an ovary into the right axillary region; (d) 12 intact male mice with subcutaneous homotransplants of an ovary into the right axillary region; (e) 9 castrated male mice with subcutaneous homotransplants of an ovary either into the right axillary region or into the abdominal region; and (f) 34 intact male mice with homo- or heterotransplants of an ovary into the right testis. Castration and grafting were done in one stage operations. Upon the removal of gonads the Fallopian tubes or sperm ducts and adipose tissues adjacent to the gonads were cauterized. The ovary was implanted into the spleen through a small incision made in the splenic capsule. The method of intratesticular transplantation of ovaries has been described (30). The ages of mice at the time of grafting varied from 1 to 3 months, except in the group (f) in which a few mice were operated upon at the age of 130 days. The donors of the ovaries for homotransplants or heterotransplants were usually younger than the recipients. The experimental animals received no treatment other than as mentioned.

Vaginal smears of the castrated female mice were taken at intervals during the experiment. Laparotomies were performed in some animals to

check for adhesions and to determine growth of the graft. At autopsy the ovarian grafts, genital tissues, adrenal glands, kidneys, submaxillary glands, and pituitary glands were examined and preserved for histological examination. Mam-

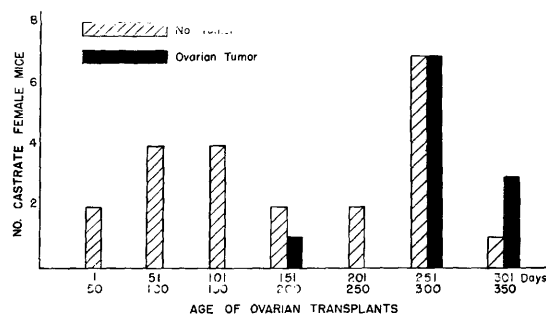


FIG. 1.—Age distribution of ovarian tumors in intrasplenic ovarian transplants in castrated male mice.

mary glands of a few mice were studied as whole mounts. Most of the ovarian grafts were sectioned serially and several areas from the ribbons were saved and stained with hematoxylin and triosin.

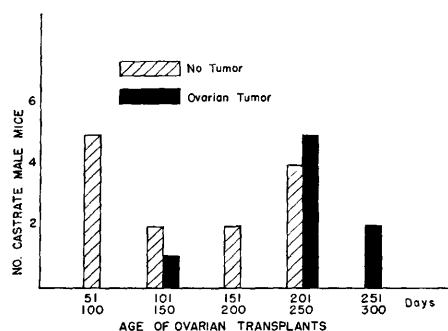


FIG. 2.—Age distribution of ovarian tumors in intrasplenic ovarian transplants in castrated female mice.

OBSERVATIONS

I. TUMORS IN INTRASPLENIC OVARIAN TRANSPLANTS

The ovarian transplants grew progressively in the spleens of castrated male and female mice and some were palpable through the abdominal wall at 3 months subsequent to grafting. At this time the transplants, observed at laparotomy, contained many large follicular cysts and they usually measured about $2 \times 4 \times 5$ mm.

The earliest granulosa-cell tumor was discovered in a castrated male mouse bearing a 130 day old graft although most of the tumors occurred in grafts over 200 days old. Most ovarian tumors were larger than the non-tumorous grafts and protruded from the surface of the spleen; the largest measured $5 \times 8 \times 9$ mm. in diameter. The tu-

¹These mice were supplied by Drs. A. Gorbman, C. W. Hooker and L. C. Strong.

²AC₃ are first generation hybrid mice of the following parentage: C3H (low-tumor) ♀ × A ♂.

AC₅ are first generation hybrid mice of the following parentage: A (low-tumor) ♀ × C3H ♂.

mors were irregular in shape, reddish yellow, and some showed a few scattered hemorrhagic cysts as indicated by dark red spots (Fig. 4).

A. Castrated male mice bearing intrasplenic ovarian transplants.—Five granulosa-cell tumors, two pretumorous lesions of the same type and one mixed tumor—both granulosa and luteoma cells—were found among the 21 castrated male mice of the A strain (Table I). With one exception these tumorous grafts were more than 200 days of age.

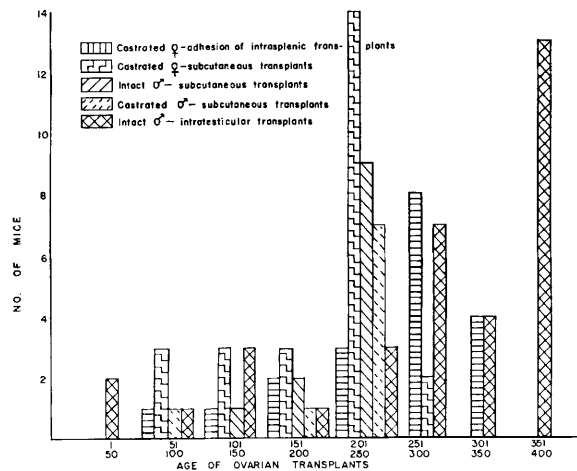


FIG. 3.—Age distribution of ovarian transplants in mice.

Among the 4 castrated male mice that survived over 200 days after the transplantation with no tumor growth only 1 showed viable ovarian tissue, a few primary follicles and some ovarian stroma constituting the small graft. The grafts in 2 other mice were composed of uterine tubal tissue or adipose tissue and in 1 animal no graft was found.

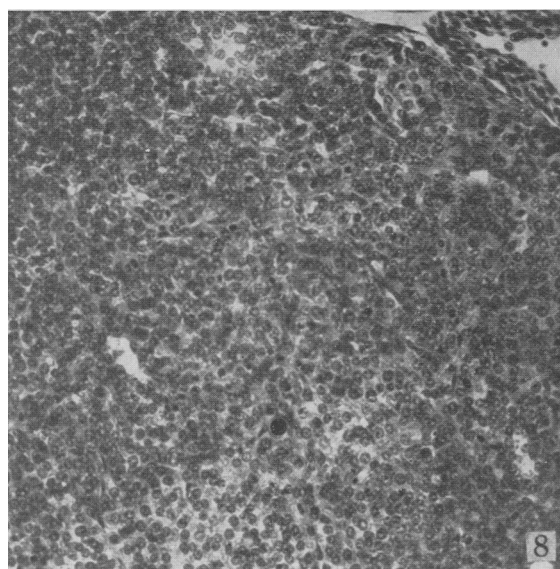
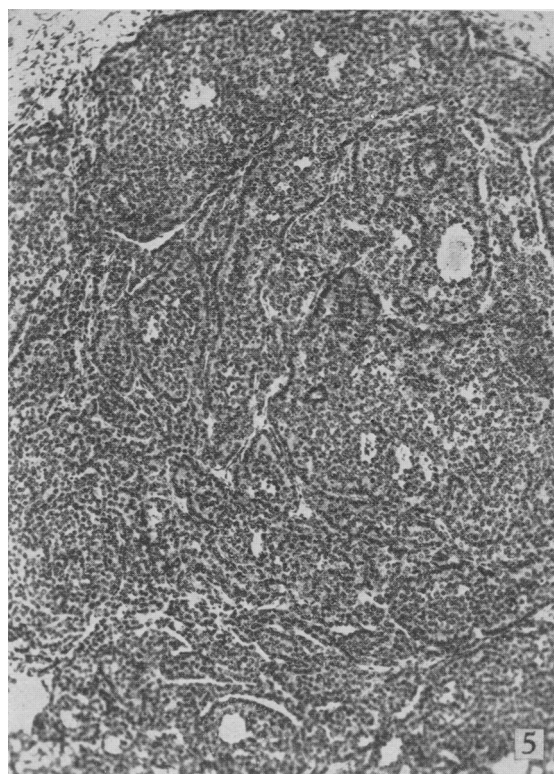
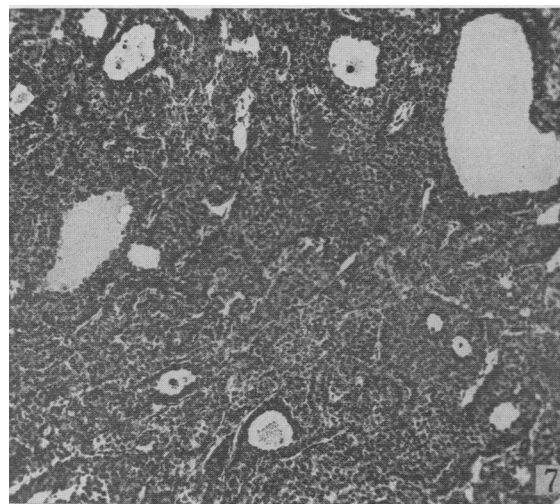
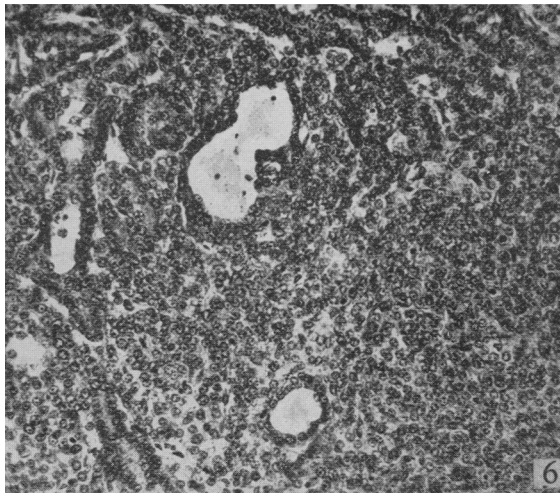
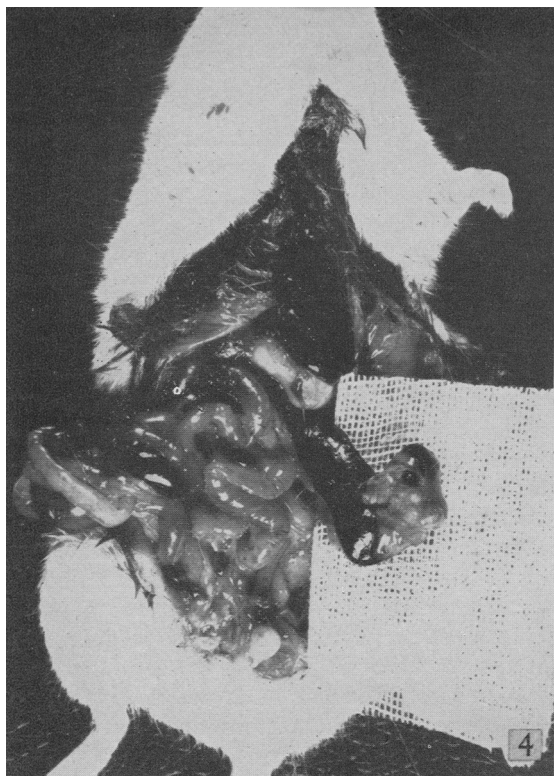
Nine mice that had no tumors of the transplanted ovaries were killed or died when the grafts were less than 200 days old. Two mice that had carried intrasplenic grafts for 170 and 198 days respectively had large transplants showing some ingrowth of the germinal epithelium and a disorganized stroma. The impression was given that, had the hosts survived for a longer period, the transplants would have become tumorous. The transplants in the two mice that had survived from 101 to 150 days were small, one could not be adequately examined because the animal had died sometime before autopsy and the other contained several large follicles and cysts of uterine tubal epithelium. Viable ovarian tissue was found in the 5 mice that had intrasplenic grafts for less than 100 days. The grafts differed in size but all contained follicles, either primary or both primary and secondary. The bulk of the grafts, however, consisted of enlarged interstitial cells with a pale

staining cytoplasm. In only 1 of the later transplants were ingrowths of the germinal epithelium noted (Fig. 1).

Microscopically, the granulosa-cell tumors consisted of cells with hyperchromatic vesicular nuclei. Most of the tumor cells were indistinguishable from the granulosa cells of normal graafian follicle; mitoses were frequent. In some areas the tumor cells invaded the splenic tissue although no metastases were observed. The tumor cells usually formed many somewhat dissociated masses and the arrangement of the cells in these masses varied considerably even in a single section of the same tumor. In some masses the tumor cells were diffuse and in others intermingled with thecal and interstitial cells; but most of the tumors contained extensive one-cell-layered cords that were more deeply stained than the adjacent tumor cells, and that were sometimes arranged in tubular structures (Fig. 5). Enlargement of the tubular spaces appeared to give rise to small and medium-sized cavities that usually contained a few degenerating cells and sometimes blood cells or coagulated fluid (Fig. 6). Also, folliculoid structure seemed to form by separation of a mass of tumor cells into smaller groups each surrounded by delicate connective tissue septa; the cavities resulted apparently by subsequent liquefaction of some of the centrally placed cells (Fig. 5, 7). A layer of spindle-shaped cells, suggesting thecal cells, was observed outside the ovarian follicle-like structures. Accumulation of the tumor cells and false ovocytes occasionally simulated the appearance of a cumulus oophorus (Fig. 7). Thus the granulosa-cell tumors were preponderantly diffuse and folliculoid in pattern as featured by Barzilai (3) in human material. Different histological arrangements of the tumor cells co-existed in the same tumor. However, two tumors referred to as the 'pretumorous' lesions were composed of small groups of diffuse granulosa cells and a few hemorrhagic cavities.

A large mass of lutein-like cells (luteoma) was observed in one of the granulosa-cell tumors, as well as many scattered lutein and interstitial cells. Some granulosa-tumor cells seemed to show transformation into lutein cells (Fig. 9). A description of luteomas will be given when the tumors in the castrated female group are described.

The proliferation and ingrowth of the germinal epithelium were observed in all these tumors and in some of the non-tumorous grafts. Local proliferation of the germinal epithelium with evidence of mitotic activity occurred frequently in the tumors, and cells with ellipsoidal nuclei simulating the cells of the germinal epithelium were often mixed with the adjacent tumor cells (Fig. 10).



FIGS. 4-8

The ingrowth of germinal epithelium formed groups of networks of tubules or cysts that sometimes contained blood cells as seen in the folliculoid structures. Many cells identified as granulosa, lutein, or interstitial cells, and primary ovocytes were observed among the network of epithelial ingrowths (Fig. 11). In some grafts the nuclei of the epithelial cells that lined the tubules appeared to become vesicular and the whole mass of epithelial ingrowth resembled the folliculoid formations (Fig. 12). An intimate connection between the epithelial ingrowths and cells of the tumors was discerned (Fig. 13).

The seminal vesicles and prostates were atrophied in the castrated male mice bearing intrasplenic ovarian grafts, and histological examination of the seminal vesicles revealed no indication of hormonal stimulation. The adrenal glands showed degeneration of the x-zone. No detectable hormonal effect on the kidneys was noted and the parietal layer of the Bowman's capsule was usually composed of squamous epithelial cells. The terminal tubules of the submaxillary glands were lined by low epithelial cells, but the percentage of total tubular and alveolar areas appeared to be equal.

B. Castrated female mice bearing intrasplenic ovarian transplants.—Four luteomas and 7 mixed tumors were found among the 33 castrated female mice of the A strain that showed no adhesion of the

intrasplenic ovarian graft to the left uterine horn or to the adjacent peritoneum. Among the 19 ovariectomized mice of the same strain with vascularized adhesions of the ovarian transplant, 1 luteoma was observed in a mouse that had irregular estrous cycles during the latter part of the experimental period (Table I). Unmixed granulosa-cell tumors were not noted. The ages of the intrasplenic ovarian transplants are summarized in Fig. 2. The mean age of the grafts that became tumorous was greater in the females (278 days) than in the males (236 days).

Microscopically, the luteomas were composed of polyhedral, lutein-like cells with finely granulated, acidophilic cytoplasm, and vesicular nuclei; mitoses were rare (Figs. 14, 15). The luteoma cells formed small tubular or cord-like structures surrounded by delicate connective tissue septa and were usually assembled, as in the granulosa-cell tumors, into large masses that showed incomplete encapsulation. Blood capillaries and sinusoidal spaces were observed among the tumor cells, and tissue mast cells and pigment-containing cells occasionally occurred in some areas. Groups of large, probably lutein cells, cells having a large lightly staining cytoplasm, were found frequently in the tumors.

Seven tumors in this group of experiments were of mixed type consisting of elements of both granulosa cells and luteomatous tissue (Fig. 16). Most

TABLE I: TUMORS IN OVARIES TRANSPLANTED TO CASTRATED MICE

Site of transplant	Hosts	Number of mice	Granulosa-cell	Luteoma	Ovarian tumors	Mixed	None
Spleen	Castrated males	21	5 (224–268 days) 2? (130, 262 days)		1 (225 days)		13 (75–268 days)
Spleen	Castrated females	33		4 (153–284 days)	7 (266–346 days)		22 (25–334 days)
Spleen (adhesion)	Castrated females	19		1 (262 days)			18 (95–346 days)
Subcutaneous	Castrated females	25	1? (243 days)				24 (61–278 days)
Subcutaneous	Intact males	12					12 (141–239 days)
Subcutaneous	Castrated males	9					9 (64–233 days)
Testis	Intact males	34					34 (32–370 days)

DESCRIPTION OF FIGURES 4 TO 8

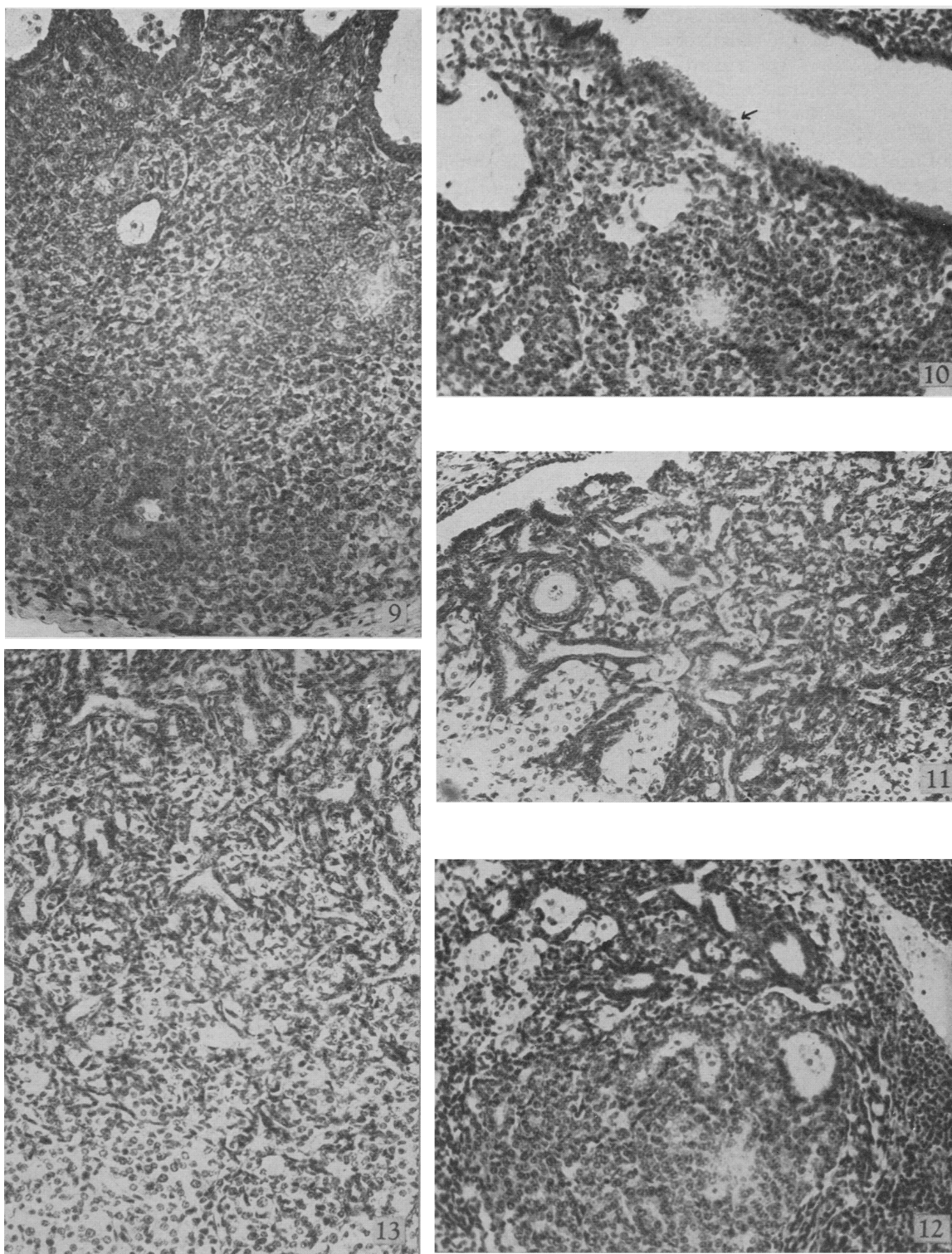
FIG. 4.—Granulosa cell tumor in an intrasplenic ovarian transplant in a castrated male mouse bearing the transplant for 268 days.

FIG. 5.—Section of a granulosa cell tumor as shown in Fig. 4. Mag. $\times 105$.

FIG. 6.—Granulosa cell tumor in an intrasplenic ovarian transplant (247 days) in a castrated male mouse, showing folliculoid structures. Mag. $\times 192$.

FIG. 7.—Folliculoid structures in a granulosa cell tumor resembling ovarian follicles. This tumor developed in an intrasplenic ovarian transplant (225 days) in a castrated male mouse. Mag. $\times 99$.

FIG. 8.—A diffuse mass of granulosa tumor cells showing mitotic figures. This tumor developed in an intrasplenic ovarian transplant (273 days) in a castrated female mouse. Mag. $\times 192$.



FIGS. 9-13

granulosa-cell tumor masses exhibited the folliculoid pattern as described above for the castrated male mice. Transformation of granulosa cells of the tumors into luteomas appeared to take place in some of the masses of the former type (Fig. 17). Many granulosa cells showed vague cell boundaries indicating that they were becoming luteinized.

Proliferation and ingrowth of the germinal epithelium were observed in all of these tumors. The luteoma cells were sometimes in direct contact with the tubular ingrowth of the germinal epithelium and sometimes intermingled at the borders of masses of less differentiated granulosa tumor cells (Fig. 18).

Four castrated female mice that had tumors of the intrasplenic ovarian transplants showed estrous vaginal smears; 2 early and 2 in the latter part of the experiment. No estrous smears were obtained from the other 8 ovariectomized mice that had tumors in the intrasplenic ovarian transplants; their uteri were small, averaging about 37 mgm. Alveolar formation was not observed in the mammary glands of two animals that had viable intrasplenic ovarian transplants. The nuclei of the stromal cells of the endometria from the castrated females with the intrasplenic ovarian tumors were generally small, dense, and spindle-shaped, resembling those found in the uteri of castrated mice (44). The x-zone of the adrenal glands was absent and areas with compact groups of rounded or fusiform cells were found in the immediate subcapsular region of the cortex of almost all adrenals from mice of this group. The latter condition was noted only in the castrated female mice. Similar changes associated with the formation of adrenal cortical carcinoma have been described by other workers in spayed female mice of the extreme dilution strain (90). As in the castrated males, the parietal layer of the renal capsules was usually composed of low epithelial cells. The relative extent of tubular and alveolar areas in the submaxillary glands were equal, but the terminal

tubules, in most cases, essentially consisted of columnar cells. The animals with adhesion of the intrasplenic ovarian transplants had submaxillary glands of the female type.

Fourteen of the 22 intrasplenic ovarian transplants that did not become tumorous were less than 250 days old, and only one tumor was noted in a younger graft (153 days). Irregular estrous smears were obtained in 12 of the 22 castrated female mice, and the average uterine weight of these 12 female hosts was 50 mgm.

The stroma of all but one of the nontumorous ovarian grafts consisted predominantly of large lutein-like interstitial cells arranged in masses, in islands separated by sinusoids, or in cords separated by sinusoids, or ingrowing strands from the germinal epithelium. Usually these cells had a granular eosinophilic cytoplasm but sometimes the cytoplasm showed little affinity for the stains. Almost all of the grafts contained a few small or primary follicles. Large follicles were found in all but one of the grafts less than 250 days old, and were rarely noted subsequently. Hemorrhagic follicles or cysts were present in all but 2 of the grafts. The first evidence of ingrowth of the germinal epithelium was noted in an ovary that had been grafted in the spleen for 110 days. None of the other 11 grafts less than 200 days old showed epithelial ingrowths whereas 5 of the 10 grafts over 200 days of age showed ingrowths of the germinal epithelium. These epithelial ingrowths were adjacent to areas of lutein-like stromal cells, and in some places the continuity was such as to suggest the origin of the latter by differentiation from the former.

No typical corpora lutea were noted but 5 of the 12 intrasplenic ovarian grafts less than 200 days old contained structures considered to be luteinized follicles, some of which closely resembled old corpora lutea. Lutein-like cells were noted along the walls of some large follicles indicating that the granulosa cells were undergoing transformation. There was no evidence of cyclic formation of

DESCRIPTION OF FIGURES 9 TO 13

FIG. 9.—Partial luteinization of granulosa tumor cells. This tumor developed in an intrasplenic ovarian transplant (247 days) in a castrated male mouse. Mag. $\times 168$.

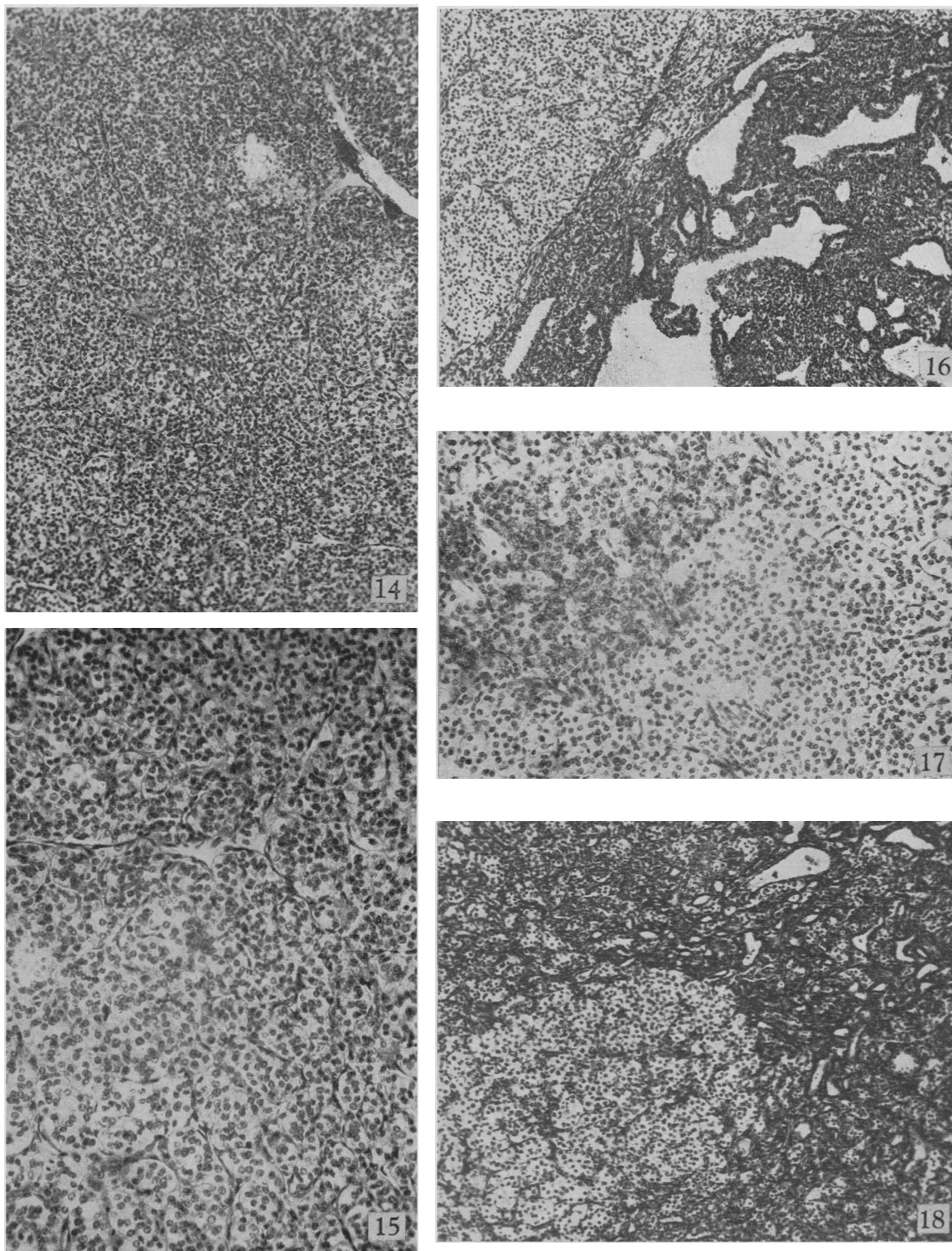
FIG. 10.—Local proliferation of germinal epithelium in an intrasplenic ovarian transplant (225 days) in a castrated male mouse. Note the mitotic activity (arrow indicated) and the close association of the germinal epithelium with granulosa tumor cells. A bursal space separated this part of transplant from the splenic tissue (above). Mag. $\times 178$.

FIG. 11.—Tubular ingrowth from germinal epithelium in an intrasplenic ovarian transplant (225 days) in a cas-

trated male mouse. Note the intimate relationship of the primary ovocyte, lutein cells and follicle cells with the network of epithelial ingrowths. Mag. $\times 171$.

FIG. 12.—Section showing the close associations of tubular ingrowths with granulosa tumor cells, and the transplant with the splenic tissue at right. Same tumor as illustrated in Fig. 10. Mag. $\times 178$.

FIG. 13.—Section showing indication of transformation of granulosa tumor cells from tubular ingrowths of the germinal epithelium. This tumor developed in an intrasplenic ovarian transplant (262 days) in a castrated male mouse. Mag. $\times 192$.



FIGS. 14-18

corpora lutea such as has been found in ovaries of intact mice. The corpora lutea-like structures were not observed in the intrasplenic grafts in the males. In this respect, in the somewhat delayed appearance of epithelial ingrowth into the stroma and of tumor formation, and in type of tumor formed, the grafts of the males and females differed appreciably.

The intrasplenic ovarian transplants in castrated female mice that had adhesion of the spleen, or graft to the adjacent body wall, uterus or kidney capsule, with one exception, were quite different from those in the mice that had no adhesion (Fig. 3). Almost all of the grafts contained a greater number of large and small ovarian follicles and corpora lutea, or follicles with completely luteinized cells (Table II). Although lutein-like stroma

and corpora lutea and a low incidence of ingrowth of the germinal epithelium.

II. SUBCUTANEOUS TRANSPLANTS OF OVARIES

A. Castrated female mice bearing subcutaneous ovarian transplants.—Among the 25 castrated female mice of the A strain with ovaries transplanted subcutaneously, one small granulosa cell-like growth was observed upon histological study (Fig. 19). The mouse was the only one that was operated upon during pregnancy and carried the graft for 243 days. Masses of granulosa cells surrounded many irregular hemorrhagic cavities and mitotic figures were frequent. Lutein cells, luteinized granulosa cells and one primary follicle were also found in this transplant. The ingrowths of the germinal epithelium formed many tubular or cystic structures that merged into the tumor-like masses. No estrous cycles were noted in this mouse during the first month after the operation, but cornified vaginal smears were obtained in the latter part of the experiment. The uterus weighed 153 mgm. at autopsy.

The other 24 castrated females carried the subcutaneous ovarian transplants for periods of 61 to 278 days. Although most of them were killed 8 months after the operation, the grafts usually were not as large or as well developed as those in the spleen, and in 7 of the 24 hosts ovarian tissues could not be definitely identified. The subcutaneous, fibro-fatty tissues encroached upon the grafts except where a capsule lined by the germinal epithelium intervened. Small masses of lymphoid tissue or areas of diffuse lymphoid tissue occupied parts of the ovarian grafts and of the adjacent tissues. Most of the ovarian grafts contained primary and maturing follicles, and luteinized follicles, some with hemorrhagic centers (Fig. 20). The stroma was composed of lutein-like interstitial cells, and in 4 grafts large areas, apparently of interstitial cells or luteinized follicles, were necrotic. Degeneration of this type has been noted in the ovaries of old mice but was not observed in the intrasplenic ovarian transplants. Ingrowth of the germinal epithelium was noted in

TABLE II: INTRASPLENIC OVARIAN TRANSPLANTS IN CASTRATED FEMALE MICE WITH ADHESIONS TO STRUCTURES DRAINED BY CAVAL SYSTEM OF VEINS*

Age group (days)	No. of mice	Number of ovarian transplants with:					
		Large follicles	Small follicles	Corpora lutea	Hemorrhagic cysts	Luteinized stroma	Germinal epithelium ingrowth
51-100	1	1	1	0	0	1	0
101-150	1	1	1	1	0	1	0
151-200	2	2	2	1	1	2	0
201-250	3	3	3	3	0	2	0
251-300	8	6	6	1	3	5	2
301-350	4	3	3	2	1	1	0

*Studies were made on test sections removed from several different areas of the transplants.

cells were apparent in several of the grafts they made up but a small part of the stroma in most instances, a more normal fibrous stroma predominating. Only in one mouse (a 262 day old transplant) was a marked overgrowth of lutein-like cells noted; they were sufficiently anaplastic in appearance to be classified as luteomatous. One of the grafts had largely regressed (270 days) and one consisted only of stroma and large sinusoids (324 days). Ingrowth of the germinal epithelium occurred in only 2 grafts, one of which was the luteoma.

In summary, the presence of adhesions was associated with the more normal constitution of the ovarian tissues, namely, the occurrence of follicles

DESCRIPTION OF FIGURES 14 TO 18

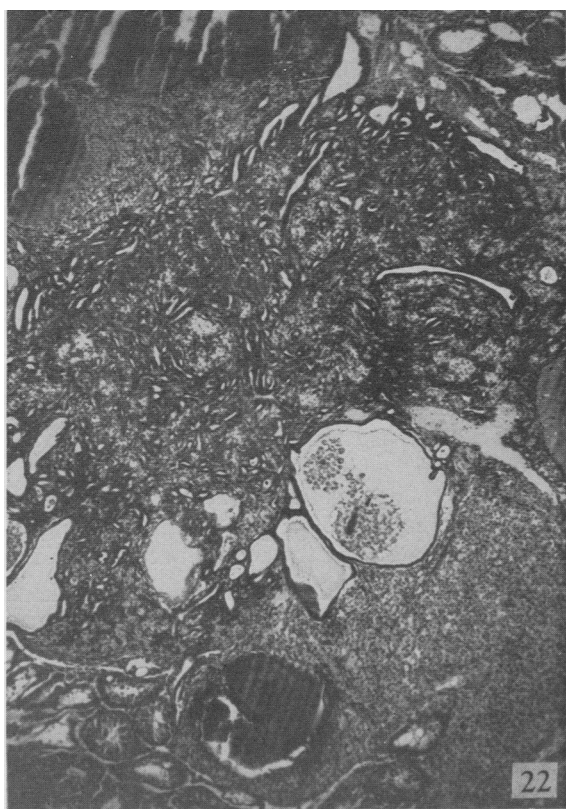
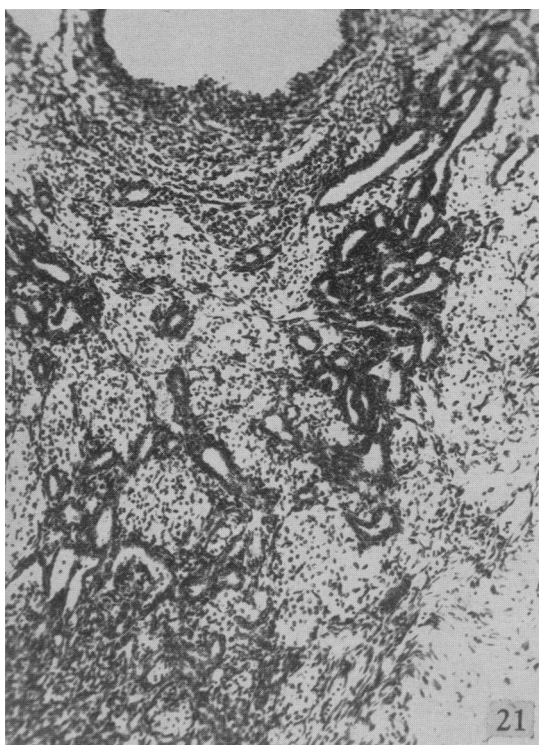
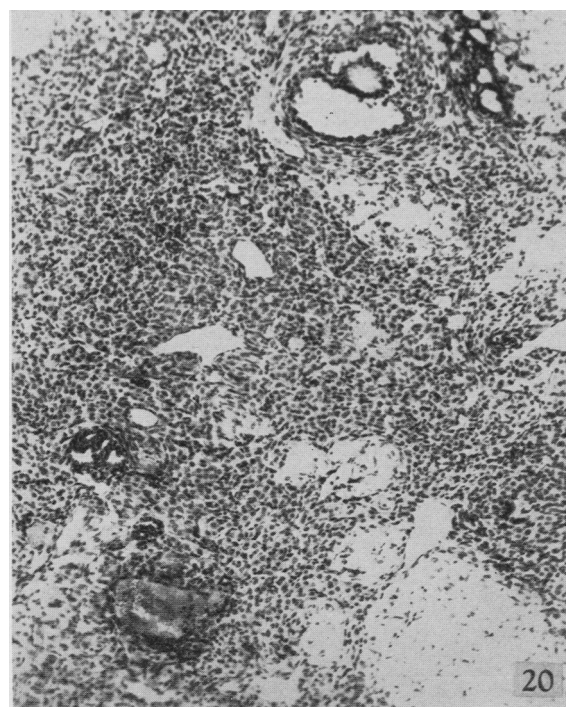
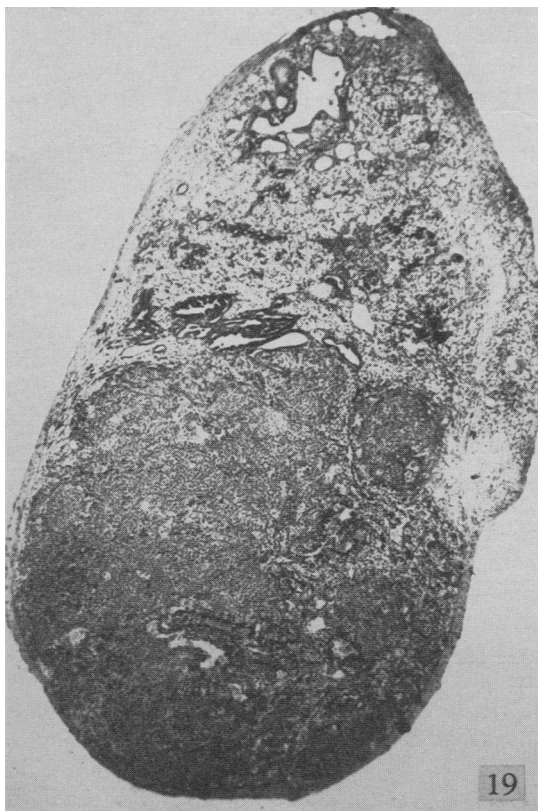
FIG. 14.—Luteoma in an intrasplenic ovarian transplant (284 days) in a castrated female mouse. Mag. $\times 93$.

FIG. 15.—Higher magnification of Fig. 14. Mag. $\times 186$.

FIG. 16.—A mixed tumor consisting of granulosa cells (right) and luteoma cells (left). This tumor developed in an intrasplenic ovarian transplant (334 days) in a castrated female mouse. Mag. $\times 89$.

FIG. 17.—Section showing indication of transformation of granulosa tumor cells into luteomatous tissue in an intrasplenic ovarian transplant (273 days) in a castrated female mouse. Mag. $\times 178$.

FIG. 18.—Section showing intimate relationship between tubular ingrowths of the germinal epithelium and luteoma cells. Same tumor as illustrated in Fig. 17. Mag. $\times 93$.



FIGS. 19-22

3 of the grafts in addition to the 1 that had become tumorous.

Estrous vaginal smears were obtained from these animals and the average weight of their uteri was 60 mgm. Some development of alveoli was maintained in the mammary glands. The adrenal glands had no x-zones and in the subcapsular region of the adrenal cortex there were compact groups of deeply stained cells similar to those observed in the group of castrated female mice with the intrasplenic ovarian transplants. Some cuboidal cells were found in the parietal layer of the Bowman's capsules. The submaxillary glands exhibited the female type of histological appearance with more alveolar than tubular structures and the terminal tubules were mainly composed of cuboidal cells.

B. Castrated male mice bearing subcutaneous ovarian transplants.—Nine castrated male mice of the A strain had ovaries placed in the subcutaneous tissue and 8 were killed 173 to 233 days later (Fig. 3). Ovarian tissue was not found in the sections prepared from 3 of the grafts; they consisted of cysts of the ovarian bursae or oviducts. Some ovarian tissue may have been present at levels other than those at which the sections were removed, although several sections were taken from different parts of each transplant.

The one transplant, removed 63 days after grafting, contained many large follicles, some cystic but still lined by granulosa cells, and several smaller follicles. Corpora lutea were not present. The ovary was largely surrounded by a bursa-like cavity. The germinal epithelium had not invaded the ovary. The 173 to 233 day old transplants from 5 mice contained a few large follicles or follicular cysts, some of which contained blood and a few small follicles. Some small follicles contained partially luteinized granulosa cells. The stroma consisted of large cells with a pale-staining cytoplasm and was broken up by extensive ingrowth of cells from the germinal epithelium (Fig. 21). In at least 2 of the ovarian transplants small areas of stroma composed of lutein-like cells were reminiscent of the intrasplenic transplants in the castrated mice, although in almost all instances the

grafts in the subcutaneous tissue were much smaller.

Atrophy of the seminal vesicles and prostates were observed in these castrated male mice. The mammary glands showed estrogenic stimulation with some growth of alveoli; and the submaxillary glands exhibited female characteristics. Groups of deeply stained small cells occurred in the subcapsular region of the adrenal cortex and no x-zone was present. In some Bowman's capsules the parietal lamina consisted partially of cuboidal epithelial cells.

C. Intact male mice bearing subcutaneous ovarian transplants.—The ovarian transplants of 8 of the 12 intact male mice bearing subcutaneous ovarian grafts for 140 to 239 days were examined histologically (Fig. 3). All of the grafts were small and were partially surrounded by bursa-like cavities. They contained few to many small and rarely a medium sized follicle. The stroma was composed primarily of compact fibroblast-like cells and "wheel cells" with small nuclei containing several aggregates of chromatin; stroma such as is found in the ovaries of hypophysectomized mice. A few "brown" cells and rarely a large cell with a clear cytoplasm were also observed in the stroma. Although the germinal epithelium was thick in many places, definite tubular ingrowths into the transplants were not noted. The small size of the grafted ovaries, absence of large follicles or corpora lutea, and the appearance of the stromal cells all indicated that the ovaries were but slightly stimulated by the gonadotrophic hormones of the hosts. In general, there were no appreciable effects of ovarian hormones in the male genital organs, adrenals, kidneys, and submaxillary glands.

III. INTRATESTICULAR TRANSPLANTS OF OVARIES

Thirty-four mice, 13 of the A strain and 21 hybrids (AC₃, AC₅), carried intratesticular ovarian grafts for 32 to 355 days. Twenty-four of the mice were killed when the grafts were over 250 days old, and 13 grafts were over 350 days old (Fig. 3).

The two grafts removed when less than 50 days old were small, were surrounded by bursa-like spaces, and contained chiefly follicles at several

DESCRIPTION OF FIGURES 19 TO 22

FIG. 19.—Granulosa cell-like growth, with evidence of epithelial ingrowth, in a subcutaneous ovarian transplant (243 days) in a castrated female mouse. Mag. $\times 42$.

FIG. 20.—Subcutaneous ovarian transplant (242 days) in a castrated female mouse showing ovocyte, tubular ingrowth, hemorrhagic cyst, interstitial cell stroma, and necrotic areas. Mag. $\times 100$.

FIG. 21.—Subcutaneous ovarian transplant (233 days) in a castrated male mouse showing follicle, tubular ingrowth, and interstitial cell stroma. Mag. $\times 100$.

FIG. 22.—Intratesticular ovarian transplant (351 days) in an intact male mouse showing follicular cysts, hemorrhagic cysts, tubular ingrowth, and interstitial cell stroma. Mag. $\times 42$.

stages of development. The older grafts were similar to one another in most respects and will not be described separately. All of the grafts, except one that was composed entirely of interstitial cells (315 days old), were covered in part by a bursa-like space or spaces lined by columnar or cuboidal epithelium, apparently the persisting germinal epithelium. Epithelial extension from the surrounding bursa occurred in most of the grafts over 250 days old and tended to increase with advancing age (Fig. 22). The epithelial extensions grew only into the ovarian tissue and not out into the testis although both sides of the bursa were lined by cells that appeared to be identical. The germinal epithelial cells did not grow into the testicular stroma. The epithelial ingrowths resembled those observed in grafts at other sites in castrated male and female hosts. In several grafts the tubular ingrowths of germinal epithelium had become cystic and sometimes contained blood. The cells lining the cysts sometimes contained hemosiderin, and in places resembled interstitial cells and macrophages.

Most of the grafts contained several follicles of different sizes. A few follicles contained blood and the cells lining the walls were somewhat luteal. No corpora lutea or completely luteinized follicles were found.

The greater part of most of the grafts consisted of a stroma composed mainly of large, lightly stained cells with definite cell membranes, and of large cells containing a brownish pigment scattered among small blood vessels and sinusoids, and fibroblastic elements. Such tissue has been classed as interstitial in the ovarian grafts at other sites.

A number of the ovarian transplants, especially the older ones, contained small areas of cord-like or solid follicle-like structures of granulosa cells somewhat reminiscent of the granulosa-cell tumors. The scarcity of mitotic figures, the small size of the areas, and well organized stroma led to the impression that they should not be designated as tumorous. The impression was gained, however, that ovarian grafts in the testes might occasionally become tumorous.

The mammary glands in the male hosts with ovarian transplants showed a partially developed duct system, although kidneys, submaxillary glands, and other genital organs showed normal male characteristics. No x-zone was present in the adrenals and the growth of compact small cells in the subcapsular region of the adrenal cortex with occasional mitotic figures was found in the hybrid male mice. In mice of the A strain, however, this condition was observed in the castrated

females and in castrated males bearing subcutaneous ovarian grafts.

DISCUSSION

I. PATHOGENESIS OF THE OVARIAN TUMORS

The high incidence of granulosa-cell tumors in women at or just preceding the menopause has been noted by several workers (15, 71, 80, 84, 87, 88), and the amenorrhea that was often ascribed to "the menopause" is now considered by Dockerty (15) as the first symptom of a granulosa-cell tumor. In presenting a case of granulosa-cell tumor in a 51 year old woman, Voigt (89) inferred that an overproduction of the gonad-stimulating hormone after the ovaries lost their inhibiting effect upon the anterior pituitary, might play a part in the formation of the tumor. Recently, Biskind and Biskind (4) reported three granulosa-cell tumors in rats' ovaries transplanted into the spleen and they suggested that the development of the tumor is due to the inactivation of ovarian estrogen by the liver, allowing the protracted stimulation of the pituitary to act on the ovary. These intrasplenic ovarian growths, however, have been questioned as "true blastomas" (80).

The present experiments provide evidence that an endocrine imbalance between the pituitary gonadotrophins and ovarian hormones is responsible for the development of granulosa-cell tumors and luteomas in mice. The increase of endogenous gonadotrophic hormones in castrated mice is manifested by the rapid growth of the ovaries and by the formation of hemorrhagic follicles shortly following the intrasplenic transplantations. The formation of hemorrhagic follicles (blood points), as Zondek (92) has pointed out, is a most conspicuous and impressive reaction in the immature mouse's ovary to the administration of anterior pituitary hormones. Furthermore, the increase of intrinsic gonadotrophins subsequent to castration has been shown by bioassays of urine, blood and hypophysis and in experimental parabiosis of intact with castrated or roentgen-rayed animals (20, 21, 23, 28, 54). Castrated female mice carrying intrasplenic ovarian transplants, but with vascularized adhesions that permitted ovarian hormones to bypass the hepatic portal circulation, showed estrous vaginal smears throughout the course of the experiment and acquired no ovarian tumors. Ovarian tumors have not occurred in the intrasplenic transplants of ovaries in unilaterally gonadectomized male and female mice (unpublished data). The only female mouse that had a granulosa cell tumor-like growth in the subcutaneously transplanted ovary was operated upon in pregnancy

and no estrous cycles were noted during the early part of the experimental period. All the other castrated male and female mice bearing ovarian transplants at sites other than those drained by the hepatic portal system, *e.g.*, subcutaneous or intratesticular grafts, had no ovarian tumors. It is not improbable that some of these grafts in the castrated mice might have become tumorous had they been observed at later dates.

It is of special interest that unmixed granulosa cell tumors developed in male, and luteomas in female, mice carrying the intrasplenic ovarian transplants. This may be due to the physiological difference in the pituitaries of males and females; such differences have been observed with rats (13, 66). The pituitary gland of males apparently produces less luteinizing hormone than that of the females. The luteinization of tumor cells and the transformation of the granulosa-like cells into luteomas indicate that the neoplastic growths are under the influence of the pituitary gonadotrophic hormones.

It seems possible that roentgen ray treatment of the ovaries in rats and mice may induce a similar endocrine imbalance. Levine and Witschi (49) reported that normal female rats showed constant estrus when placed in parabiosis with irradiated female rats, and the castration-cells were found in the pituitaries of the irradiated rats. Experimental induction of ovarian tumors in mice subsequent to roentgen ray irradiation has been reported by several workers. Because the histological structure of granulosa-cell tumors and the luteomas appearing in the ovaries grafted into the spleen are similar, if not identical, to the ovarian tumors induced with roentgen rays³, it seems that the same mechanism, namely gonadotrophic overaction, is responsible for the tumors under both circumstances.

In the present experiments ovarian tumors occurred in mice of the A strain. The question of strain limitation in formation of the tumors in mice may be raised. Results of experiments in progress show rapid growth of the intrasplenic ovarian transplants in mice of several strains. Lipschütz and his collaborators (52) mentioned that species differences and the duration of the experiment may account for failure to obtain malignant ovarian growths in guinea pigs. Only a few of their animals carried intrasplenic ovarian grafts for more than 1 year. However, most of the ovarian tumors in the experiments with mice developed 7 months after the transplantation.

³Dr. H. S. Kaplan was kind enough to permit examination of some of his unpublished material of roentgen-ray induced ovarian tumors in mice.

One testicular tumor "bearing a striking resemblance to the granulosa-cell tumor of the ovary" and with certain portions showing arrhenoblastoma-like structures developed in an intrasplenic testicular transplant in a castrated male rat (5). Intrasplenic testicular transplants in castrated male and female mice have not become tumorous (51). The higher temperature in the abdominal cavity may prevent the development of testicular transplants (59).

II. HISTOGENESIS OF THE OVARIAN TUMORS

Although the development of granulosa-cell tumors and luteomas has been studied by many investigators, the histogenesis of these tumors remains uncertain. The main controversial opinions on the histogenesis of granulosa-cell tumors reflect to a great extent the conflicting concepts of the development of definitive germ cells in the ovary. An adequate study of the histogenesis of the ovarian tumors has often been difficult because the comparatively large size of these growths precluded a systematic study of serial sections and frequently only late stages were obtained.

The histogenesis of granulosa-cell tumors has been explained by three major theories: (A) origin from differentiated granulosa cells, (B) origin from embryonic rests, and (C) origin from mesenchyme. Histological study of the present material suggests another view of the genesis of ovarian tumors, namely, that the tumors arise from the proliferation and ingrowth of germinal epithelium. The three theories may be briefly summarized as follows:

(A) Origin from differentiated granulosa cells. Schröder (76) designated the term "folliculoma" for some granulosa-cell tumors and stated that the tumors arose from the follicular epithelium. Robinson (72) reported that ova were present in the folliculoid spaces, and that the follicle cells could undergo structural and quantitative variations of the widest range and could give rise to benign and malignant tumors. He contended that the folliculoid structures in the tumors were not the result of a process of "liquefaction" or of "degeneration", not even of a possible "reversion to type" by the granulosa cells, but were the remains of ovarian follicles. Moreover, he believed that in man there is no postnatal ovogenesis from the surface epithelium of the ovary. Dockerty and MacCarty (16) described one granulosa-cell neoplasm and supported the view that the tumor arose from the follicle cells. It has been reported that in mice roentgen irradiation destroyed ova, but left the epithelium of small follicles intact (12, 29, 85) and that granulosa cell tumors originated from a proliferation of the persisting follicle cells. Others

have maintained that the ovarian tumors produced in irradiated mice were derived from the undifferentiated parenchymal cells of the ovary, and that the granulosa cells of the graafian follicle played no role (34). They observed, however, that the theca interna cells might participate with the parenchymal cells in the early postirradiation proliferation and luteinization.

(B) Origin from embryonic rests. According to Meyer (57, 58) the tumor cells originate from redundant granulosa cells that dip down into the ovarian substance from the germinal epithelium during the formative period and remain undifferentiated. These embryonic rests under certain unknown conditions become granulosa-cell tumors. Meyer based his theory upon an isolated finding of small masses of granulosa-cells in the ovary of a 45 year old woman, and assumed that such cells were situated chiefly in the medullary portion of the ovary. He contended that granulosa cell tumors cannot arise from epithelial cells of the true follicle because the life of the well differentiated granulosa cells is dependent upon living ova, and the majority of the tumors occur in old women in whom the ovaries no longer contain follicles. Te Linde (84) described one small granulosa-cell tumor located in the medulla near the hilus of the ovary, and assumed that the only possible origin of the tumor was an embryonic granulosa rest. Similar opinions have been expressed by others (32, 62).

(C) Origin from mesenchyme. Fischel (27) suggested that the ovarian stroma and follicular cells are derived from the ovarian mesenchyme, and the primitive mesenchymal cells could assume epithelial properties under the influence of the germ cells. He stated that the germ cells migrate into the gonad from the primordial gut during early fetal life, and the germinal epithelium, therefore, did not give rise to ova, follicle cells, or connective tissue, but existed throughout life merely as the peritoneal covering of the ovary. According to Fischel's theory, tumor cells of both granulosa and theca types arise through metamorphosis of the ovarian mesenchyme. Schiller (73, 74) supported this theory and stated that the transformation of granulosa-cell tumors from an immature trabecular "form" into a mature folliculoid "form" resulted from the transformation of connective tissue-like cells into the epithelial tumor cells. This mesenchymal theory has been used as a working hypothesis by several workers (37, 63, 64, 89). A participation in the histogenesis of granulosa-cell tumors also has been suggested (87).

The present experiments provided evidences that the germinal epithelium of the engrafted

mouse's ovary proliferates and gives rise to ova and follicle cells to form primary ovarian follicles. The proliferation and ingrowth of the germinal epithelium were observed in all ovarian transplants that developed into tumors and in many of the nontumorous grafts in the intrasplenic, intratesticular, and subcutaneous sites. Local proliferation of the germinal epithelium with evidence of mitotic activity occurred frequently in the tumors. The intermingling of the invaginated epithelial cells with the cells of the smaller granulosa-cell tumors and the apparent transformation of the epithelial cells, with ellipsoidal nuclei and scanty cytoplasm, into the tumor cells indicated the genesis of granulosa-cell tumors from the germinal epithelium.

The ingrowths of germinal epithelium formed groups or networks of tubules or cysts. In some instances nuclei of the epithelial cells that lined the tubules appeared to become vesicular and the whole mass of epithelial ingrowth resembled the folliculoid formations. More often, however, the folliculoid structure seemed to form within small masses of diffuse tumor cells by subsequent liquefaction of some of the centrally placed cells. Schiller (73, 74) appeared to have noted transformations of this type although he did not associate the tumor formation with the activity of the germinal epithelium in his materials. Many of the follicle-like structures contained a few degenerating cells and sometimes blood cells or coagulated fluid, but no ovocytes were found in the folliculoid spaces in the present materials. Small ovarian follicles were rarely encountered at the periphery of ovarian transplants that developed into tumors. In the nontumorous ovarian transplants the number of ovarian follicles was generally decreased in proportion to the duration of the graft. There was no indication that the granulosa tumor cells were derived from the well differentiated granulosa or theca cells of the persisting ovarian follicles.

The observations made in the present study, as well as the histological examinations of ovaries of mice in other experiments in our laboratories, have not revealed embryonic rests of granulosa cells in the ovaries. Our experimental results do not support the assumption that such an embryonic rest of granulosa cells is a prerequisite condition for tumor formation. The tumors did not appear to develop in the medullary portion of the engrafted ovary but arose in conjunction with the germinal epithelium or in conjunction with ingrowths from the germinal epithelium. The proliferation and ingrowth of the germinal epithelium were not restricted to any particular region of the ovary.

Despite the discrepancy regarding the histo-

genesis of granulosa-cell tumors, pathologists agree that some ovarian tumors, *e.g.*, cystadenomas and adenocarcinomas, are derived from the germinal epithelium (32, 56, 72). The ingrowth of the germinal epithelium in roentgen-rayed ovaries of immature and mature mice, and rats has been described (18, 61, 66). Other workers (12, 29, 85) observed "tubular adenomas," but did not report a relationship between the invaginated epithelial cells and the granulosa-cell tumors and luteomas. However, the so-called tubular adenomas in roentgen-rayed ovaries of mice differ from tubular adenomas in the ovaries of old rats (22).

The fundamental issue in the histogenesis of granulosa-cell tumors is more closely associated with the development of ovarian follicle cells than the germ cells, although they have an intimate relationship. Allen (1) called attention to the cyclical production of new germ cells from the germinal epithelium during each estrous cycle in the adult female mouse. This observation on rhythmic activity of the germinal epithelium in mature female mice has been substantiated by several other workers (8, 9). Recently, Long (53) reported the growth *in vitro* of ovarian germinal epithelium from the adult mouse's ovary and observed the formation of primary follicles from the epithelial cells. The origin of definitive ova from the germinal epithelium in the ovaries of adult rats has also been studied (11, 24, 39, 48). Hargitt (39) described the production of ova, follicle cells, and interstitial cells from the germinal epithelium in adult rats' ovaries, and in old and senile rats the germinal epithelium continues to be active, especially in producing invaginated cords of cells, but frequently they are anovular. In an extensive investigation, Evans and Swezy (24) found that ovogenesis in the rat, guinea pig, dog, cat and man occurs throughout the whole period of sexual life in a rhythm fundamentally related to the ovulation cycle. The ova and follicle cells arise by proliferations from the germinal epithelium in the form of invaginations or cords. Hartman (41) and Everett (25) reported that in the adult opossum the germinal epithelium near the region of the hilus is more active in producing new ova; and Papanicolaou (65) suggested the increased activity in the hilus of the guinea pig's ovary was due to a greater supply of blood. Early studies on the continual proliferation of germ cells from the germinal epithelium have been reviewed (83). There are many reports supporting the view that the germinal epithelium is active in producing ova in adult mammalian ovaries: the bat (38), the guinea pig (7, 40, 75), the rabbit (19), the ground squirrel (70), the prairie dog (82) and the dog (2).

Luteomas appear to arise also from the tubular ingrowths of the germinal epithelium, although there are instances showing that the tumors may arise indirectly by transformation from granulosa tumor cells. The cells in a granulosa-cell tumor may partially luteinize, and large lutein cells are present among the luteomatous cells. The possibility that luteomas originate from proliferation of the lutein cells, as suggested by Butterworth (12), was not confirmed by our observations. Luteinization of the granulosa tumor cells has been reported (43, 63, 86). The present studies reveal that granulosa-cell tumors and luteomas are derived from the germinal epithelium. The general concept of ovarian epithelial tumors originated from the invading germinal epithelium has been postulated by Goodall (36) and others.

SUMMARY AND CONCLUSIONS

1. Five granulosa-cell tumors, 2 pretumorous lesions and 1 mixed tumor (granulosa and luteoma cells) were found among 21 castrated male mice of the A strain bearing intrasplenic ovarian transplants. Five luteomas and 7 mixed tumors occurred in intrasplenic ovarian transplants in 52 castrated female mice of the same strain. Most of the ovarian tumors developed after 7 months from time of operation.

2. Castrated female mice bearing intrasplenic ovarian transplants with vascularized adhesions that permitted drainage through other than the hepatic portal system showed estrous vaginal smears throughout the course of the experiment and acquired no ovarian tumors. The only female mouse that had a granulosa cell tumor-like growth in the subcutaneously transplanted ovary was operated upon during pregnancy and no estrous cycles were noted during the early part of the experimental period. All the other intact or castrated male and female mice bearing ovarian transplants in subcutaneous or testicular sites had no ovarian tumors.

3. The intrasplenic ovarian transplants in the gonadectomized hosts attained larger size and more frequently showed extensive tubular ingrowth of germinal epithelium than did those at other sites.

4. The germinal epithelium in the ovarian transplants give rise to the primary ovocytes and follicle cells. The granulosa-cell tumors appear to be derived from the ingrowth of germinal epithelial cells.

5. The luteomas develop either directly from the ingrowth of the germinal epithelium or indirectly from granulosa-cell tumors.

6. Ovarian hormones produced by the intra-splenic ovarian transplants are inactivated by the liver before entering the systemic circulation, while the hypophysis, under such conditions as physiological castration, produces larger amounts of gonadotrophic hormones to further stimulate the ovarian transplants. The endocrine imbalance so established is assumed to be responsible for the genesis of the ovarian tumors in the present experiments and possibly also in the ovaries of roentgen ray irradiated mice.

7. The influence of ovarian transplants on the accessory genital tissues and other organs in mice was described.

REFERENCES

1. ALLEN, E. Ovogenesis during Sexual Maturity. *Am. J. Anat.*, **31**:439-482. 1923.
2. BARTON, E. P. The Cyclic Changes of Epithelial Cords in the Dog Ovary. *J. Morphol.*, **77**:317-349. 1945.
3. BARZILAI, G. Atlas of Ovarian Tumors. New York: Grune & Stratton. 1943.
4. BISKIND, M. S., and BISKIND, G. R. Development of Tumors in the Rat Ovary after Transplantation into the Spleen. *Proc. Soc. Exper. Biol. & Med.*, **55**:176-179. 1944.
5. BISKIND, M. S., and BISKIND, G. R. Tumor of Rat Testis Produced by Heterotransplantation of Infantile Testis to Spleen of Adult Castrate. *Proc. Soc. Exper. Biol. & Med.*, **59**:4-8. 1945.
6. BISKIND, G. R., and MARK, J. The Inactivation of Testosterone Propionate and Estrone in Rats. *Bull. Johns Hopkins Hosp.*, **65**:212-217. 1939.
7. BOOKHOUT, C. G. The Development of the Guinea Pig Ovary from Sexual Differentiation to Maturity. *J. Morphol.*, **77**:233-263. 1945.
8. BRAMBELL, F. W. R. The Development and Morphology of the Gonads of the Mouse. I. The Morphogenesis of the Indifferent Gonad and of the Ovary. *Proc. Roy. Soc. (S. B.)*, **101**:391-408. 1927.
9. BULLOUGH, W. S. Oogenesis and Its Relation to the Oestrous Cycle in the Adult Mouse. *J. Endocrinology*, **3**:141-149. 1942.
10. BURROWS, H. Biological Actions of Sex Hormones. London: Cambridge Univ. Press. 1945.
11. BUTCHER, E. O. The Origin of Definitive Ova in the White Rat (*Mus Norvegicus Albinus*). *Anat. Rec.*, **37**:13-29. 1927.
12. BUTTERWORTH, J. S. Observation on the Histogenesis of Ovarian Tumors Produced in Mice by X-Rays. *Am. J. Cancer*, **31**:85-99. 1937.
13. CLARK, HELEN M. A Prepubertal Reversal of the Sex Difference in the Gonadotropic Hormone Content of the Pituitary Gland of the Rat. *Anat. Rec.*, **61**:175-192. 1935.
14. CRABTREE, C. E. The Structure of Bowman's Capsule in Castrate and Testosterone Treated Male Mice as an Index of Hormonal Effects on the Renal Cortex. *Endocrinology*, **29**:197-203. 1941.
15. DOCKERTY, M. B. Ovarian Neoplasms, A Collective Review of the Recent Literature. *Internat. Abstr. Surg.*, **81**:179-204. 1945.
16. DOCKERTY, M. B., and MACCARTY, W. C. SR., Granulosa Cell Neoplasm with a Discussion of Possible Histogenesis. *Am. J. Obst. & Gynec.*, **38**:698-702. 1939.
17. DOSNE, CHRISTIANA, Inactivation of Antifibromatogenic Substances (Progesterone and Desoxycorticosterone Acetate) in the Liver. *Cancer Research*, **4**:512-514. 1944.
18. DRIPS, D. G., and FORD, FRANCES A. The Study of the Effects of Roentgen Rays on the Estrual Cycle and the Ovaries of the White Rat. *Surg. Gynec. & Obst.*, **55**:596-606. 1932.
19. DUKE, K. L. The Germ Cells of the Rabbit Ovary from Sex Differentiation to Maturity. *J. Morphol.*, **69**:51-81. 1941.
20. DUSHANE, G. P., LEVINE, W. T., PFEIFFER, C. A., and WITSCHI, E. Experimental "Constant Oestrus" and the Notion of Antigonadotropic Hormones. *Proc. Soc. Exper. Biol. & Med.*, **33**:339-345. 1935.
21. ENGLE, E. T. The Effect of Daily Transplants of the Anterior Lobe from Gonadectomized Rats on Immature Test Animals. *Am. J. Physiol.*, **88**:101-106. 1929.
22. ENGLE, E. T. Tubular Adenomas and Testis-like Tubules of the Ovaries of Aged Rats. *Cancer Research*, **6**:578-582. 1946.
23. EVANS, H. M., and SIMPSON, MIRIAM E. A Comparison of Anterior Hypophyseal Implants from Normal and Gonadectomized Animals with Reference to Their Capacity to Stimulate the Immature Ovary. *Am. J. Physiol.*, **89**:371-374. 1929.
24. EVANS, H. M., and SWEZY, O. Ovogenesis and the Normal Follicular Cycle in Adult Mammalia. *Mem. Univ. Calif.*, **9**:119-224. 1931.
25. EVERETT, N. B. The Origin of Ova in the Adult Opossum. *Anat. Rec.*, **82**:77-91. 1942.
26. FEKETE, E. Chapter 3 in *Biology of the Laboratory Mouse*, Ed. by Snell, Philadelphia: G. D. Blakiston. 1941.
27. FISCHER, A. Ueber die Entwicklung der Keimdrüsen des Menschen. *Ztschr. f. d. ges. Anat. (Abt. 1)*, **92**:34-72. 1930.
28. FLUHMAN, C. F., and MURPHY, K. M. Estrogenic and Gonadotropic Hormones in the Blood of Climacteric Women and Castrates. *Am. J. Obst. & Gynec.*, **38**:778-785. 1939.
29. FURTH, J., and BUTTERWORTH, J. S. Neoplastic Diseases Occurring Among Mice Subjected to General Irradiation with X-rays. II. Ovarian Tumors and Associated Lesions. *Am. J. Cancer*, **28**:66-95. 1936.
30. GARDNER, W. U. The Effect of Ovarian Hormones and Ovarian Grafts upon the Mammary Glands of Male Mice. *Endocrinology*, **16**:656-667. 1935.
31. GARDNER, W. U. Tumors in Experimental Animals Receiving Steroid Hormones. *Surgery*, **16**:8-32. 1944.
32. GEIST, S. H. Histogenesis of Certain Ovarian Tumors, and Their Biologic Effects. *Am. J. Obst. & Gynec.*, **30**:650-664. 1935.
33. GEIST, S. H. Ovarian Tumors. New York: P. B. Hoeber. 1942.
34. GEIST, S. H., GAINES, J. A., and POLLACK, A. D. Experimental Biologically Active Ovarian Tumors in Mice. Histogenesis and Relationship to Similar Human Ovarian Tumors. *Am. J. Obst. & Gynec.*, **38**:786-797. 1939.
35. GOLDEN, JUNE B., and SEVRINGHAUS, E. L. Inactivation of Estrogenic Hormone of the Ovary by the Liver. *Proc. Soc. Exper. Biol. & Med.*, **39**:361-362. 1938.

36. GOODALL, J. R. The Origin of Tumors of the Ovary. *Surg. Gynec. & Obst.*, **30**:249-264. 1920.
37. GREENHILL, J. P., and GREENBLATT, R. B. Status of the Thecoma and its Relationship to the Granulosa Cell Tumor. *Am. J. Obst. & Gynec.*, **36**:685-688. 1938.
38. GUTHRIE, M. J., and JEFFERS, K. R. A Cytological Study of the Ovaries of Bats, *Myotis lucifugus lucifugus* and *Myotis grisescens*. *J. Morphol.*, **62**:523-556. 1938.
39. HARGITT, G. T. The Formation of the Sex Glands and Germ Cells of Mammals. V. Germ Cells in the Ovaries of Adult, Pregnant, and Senile Albino Rats. *J. Morphol.*, **50**:453-469. 1930.
40. HARMAN, M. T., and KIRGIS, H. D. The Development and Atresia of the Graafian Follicle and the Division of Intraovarian Ova in the Guinea Pig. *Am. J. Anat.*, **63**:79-99. 1938.
41. HARTMAN, C. G. Postpubertal Oogenesis in the Opossum. *Anat. Rec.*, **32**: (Supp.) 209 1926. Abstr., *Am. Assoc. Anatomists*, 42nd Ann. Session, New Haven, April 1, 1926.
42. HELLBAUM, A. A., and GREEP, R. O. Qualitative Changes Induced in Gonadotropic Complex of Pituitary by Testosterone Propionate. *Endocrinology*, **32**:33-40. 1943.
43. HENDERSON, D. N. Granulosa and Theca Cell Tumors of the Ovary. *Am. J. Obst. & Gynec.*, **43**:194-210. 1942.
44. HOOKER, C. W. A Criterion of Luteal Activity in the Mouse. *Anat. Rec.*, **93**:333-347. 1945.
45. ISRAEL, S. L., MERANZE, D. R., and JOHNSTON, C. G. The Inactivation of Estrogen by the Liver. Observations on the Fate of Estrogen in Heart-Lung and Heart-Lung-Liver Perfusion Systems. *Am. J. M. Sc.*, **194**:835-843. 1937.
46. KOCHAKIAN, C. D., HASKINS, A. L. JR., and BRUCE, R. A. The Site of Metabolism of Progesterone in the Rabbit. *Am. J. Physiol.*, **142**:326-327. 1944.
47. LACASSAGNE, A. Dimorphisme sexual de la glande sous-maxillaire chez la souris. *Compt. rend. de Soc. biol.*, **133**:180-181. 1940.
48. LATTA, J. S., and PEDERSON, E. S. The Origin of Ova and Follicle Cells from the Germinal Epithelium of the Ovary of the Albino Rat as Demonstrated by Selective Intravital Staining with India Ink. *Anat. Rec.*, **90**:23-25. 1944.
49. LEVINE, W. T., and WITSCHI, E. Endocrine Reactions in Female Rats after X-Ray Treatment of the Ovaries. *Proc. Soc. Exper. Biol. & Med.*, **30**:1152-1153. 1933.
50. LI, M. H., and GARDNER, W. U. Tumors in Intrasplenic Ovarian Transplants in Castrated Mice. *Science*, **105**:13-15. 1947.
51. LI, M. H., PFEIFFER, C. A., and GARDNER, W. U. Intrasplenic Transplantation of Testis in Castrated Mice. *Proc. Soc. Exper. Biol. & Med.*, **64**:319-323. 1947.
52. LIPSCHUTZ, A., DE LEON, H. P., WOYWOOD, E., and GAY, O. Intrasplenic Ovarian Grafts in the Guinea Pig and the Problem of Neoplastic Reactions of the Graft. *Rev. Canad. Biol.*, **5**:181-198. 1946.
53. LONG, J. A. Growth in vitro of Ovarian Germinal Epithelium. *Contrib. Embryol., Carnegie Inst.*, **172**:89-96. 1940.
54. MARTINS, T. Influence de l'épithélium séminal sur l'hypophyse (Expérience de parabiose.) *Compt. rend. Soc. Biol.*, **105**:789-790. 1930.
55. MASSON, G., and HOFFMAN, M. M. Studies on the Role of the Liver in the Metabolism of Progesterone. *Endocrinology*, **37**:111-116. 1945.
56. MEYER, R. Zur Histogenese und Einteilung der Ovarialkystome. Eine kritische Literatursichtung. *Monatschr. f. Geburtsh. u. Gynäk.*, **44**:302-331 1916.
57. MEYER, R. Drei Beiträge zur Kenntnis seltenerer Ovarialtumoren. *Arch. f. Gynäk.*, **109**:212-246. 1918.
58. MEYER, R. The Pathology of Some Special Ovarian Tumors and Their Relation to Sex Characteristics. *Am. J. Obst. & Gynec.*, **22**:697-713. 1931.
59. MOORE, C. R. Biology of the Testis in "Sex and Internal Secretions." Second edition. Baltimore: Williams & Wilkins, 1939.
60. MOORE, C. R., and PRICE, DOROTHY. Gonad Hormone Functions, and the Reciprocal Influence between Gonads and Hypophysis with Its Bearing on the Problem of Sex Hormone Antagonism. *Am. J. Anat.*, **50**:13-71. 1932.
61. MURRAY, J. M. A study of the Histological Structure of Mouse Ovaries Following Exposure to Roentgen Irradiation. *Am. J. Roentgenol.*, **25**:1-45. 1931.
62. NOVAK, E. Granulosa Cell Carcinoma of Ovary as a Cause of Postmenopausal Bleeding. *Am. J. Surg.*, **24**:595-609. 1934.
63. NOVAK, E. Ovarian Tumors of Endocrine Nature. *J. A. M. A.*, **116**:947-950. 1941.
64. NOVAK, E. Endocrine Effects of Certain Dysontogenetic Tumors of the Ovary. *Endocrinology*, **30**:953-958. 1942.
65. PAPANICOLAOU, G. N. Ovogenesis during Sexual Maturity as Elucidated by Experimental Methods. *Proc. Soc. Exper. Biol. & Med.*, **21**:393-396. 1942.
66. PARKS, A. S. The Internal Secretion of the Ovary. London: Longmans, Green & Co. 1929.
67. PFEIFFER, C. A. Sexual Differences of the Hypophyses and their Determination by the Gonads. *Am. J. Anat.*, **58**:195-225. 1936.
68. PFEIFFER, C. A., EMMEL, V. M., and GARDNER, W. U. Renal Hypertrophy in Mice Receiving Estrogens and Androgens. *Yale J. Biol. & Med.*, **12**:493-501. 1940.
69. PFEIFFER, C. A., and HOOKER, C. W. Testicular Changes Resembling Early Stages in the Development of Interstitial Cell Tumors in Mice of the A Strain after Long-Continued Injections of Pregnant Mare Serum. *Cancer Research*, **3**:762-766. 1943.
70. PLISKE, E. C. The Follicular Cycle in the Sexually Mature 13-Lined Ground Squirrel (*Citellus tridecemlineatus* Mitch). *J. Morphol.*, **63**:263-287. 1938.
71. PRATT, F. B. Granulosa-Cell Tumors of the Ovary. Review of the Literature. *J. Obst. & Gynec., Brit. Emp.*, **44**: 880-933. 1937.
72. ROBINSON, M. R. Primary and Secondary Ovarian Cancer. A Histogenetic, Morphological and Clinical Study. *Surg. Gynec. & Obst.*, **51**:321-344. 1930.
73. SCHILLER, W. Pathologie und der Klinik Granulosazell-tumoren. *Mein: Wilhelm Maudrich*. 1934.
74. SCHILLER, W. Recent Findings in Solid Ovarian Tumours. *J. Obst. & Gynec., Brit. Emp.*, **43**:1135-1144. 1936.
75. SCHMIDT, IDA G., and HOFFMAN, F. G. Proliferation and Ovogenesis in the Germinal Epithelium of the Normal Mature Guinea Pig Ovary, as Shown by the Colchicine Technique. *Am. J. Anat.*, **68**:263-273. 1941.

76. SCHRÖDER, H. Ueber das Vorkommen von Follikelanlagen in Neubildungen. Ein Beitrag zur Entstehung der Eierstockgeschwülste. *Arch. f. Gynäk.*, **64**:193-236. 1901.
77. SEGALOFF, A., and NELSON, W. O. Absorption and Metabolism of Estrogens. I. Estrogen Assay by Intrasplenic Injection. *Proc. Soc. Exper. Biol. & Med.*, **48**:33-34. 1941.
78. SELYE, H. On the Role of the Liver in the Detoxification of Steroid Hormones and Artificial Estrogens. *J. Pharmacol. & Exper. Therap.*, **71**:236-238. 1941.
79. SELYE, H. Experimental Investigations Concerning the Role of the Pituitary in Tumorigenesis. *Surgery*, **16**:33-64. 1944.
80. SELYE, H. Ovarian Tumors. *Encyclopedia of Endocrinology*, Sect. IV, Vol. VII. Montreal: Richardson, Bond and Wright. 1946.
81. SMITH, P. E., and ENGLE, E. T. Experimental Evidence Regarding the Role of the Anterior Pituitary in the Development and Regulation of the Genital System. *Am. J. Anat.*, **40**:159-217. 1927.
82. STOCKARD, A. H. Studies on the Female Reproductive System of the Prairie Dog, *Cynomys leucurus*. II. Normal Cyclic Phenomena of the Ovarian Follicles. *Pap. Mich. Acad. Sci. Arts & Letters*, **22**: 671-690. 1936.
83. SWEZY, O. The Changing Concept of Ovarian Rhythms. *Quart. Rev. Biol.*, **8**:423-433. 1933.
84. TELINDE, R. W. Granulosa-cell Tumors of the Ovary and their Relation to Post-menopausal Bleeding. *Am. J. Obst. & Gynec.*, **20**:552-570. 1930.
85. TRAUT, H. F., and BUTTERWORTH, J. S. The Theca, Granulosa, Lutein Cell Tumors of the Human Ovary and Similar Tumors of the Mouse's Ovary. *Am. J. Obst. & Gynec.*, **34**:987-1006. 1937.
86. TRAUT, H. F., KUDER, A., and CADDEN, J. F. A Study of the Reticulum and of Luteinization in Granulosa and Theca Cell Tumors of the Ovary. *Am. J. Obst. & Gynec.*, **38**:798-814. 1939.
87. TRAUT, H. F., and MARCHETTI, A. A. A Consideration of So Called "Granulosa" and "Theca" Cell Tumors of the Ovary. *Surg. Gynec. & Obst.*, **70**: 632-642. 1940.
88. VARANGOT, J. Les Tumeurs de la Granulosa (Folliculomes de l'Ovaire). Paris: Louis Arnette. 1937.
89. VOIGT, W. W. Primary Giant Granulosa Cell Tumor of Retroperitoneal Origin with Development into the Mesosigmoideum. *Am. J. Obst. & Gynec.*, **36**: 688-693. 1938.
90. WOOLLEY, G. W., and LITTLE, C. C. The Incidence of Adrenal Cortical Carcinoma in Gonadectomized Female Mice of the Extreme Dilution Strain. I. Observations on the Adrenal Cortex. *Cancer Research*, **5**:193-202. 1945.
91. ZONDEK, B. Ueber das Schicksal des Follikelhormons (Follikulin) im Organismus. *Skandinav. Arch. Physiol.*, **70**:133-167. 1934.
92. ZONDEK, B. Clinical and Experimental Investigations on the Genital Functions and their Hormonal Regulation. Baltimore: Williams & Wilkins. 1941.
93. ZONDEK, B., and ASCHHEIM, S. Das Hormone des Hypophysenvorderlappens. *Klin. Wchenschr.*, **6**: 248-252. 1927.